PARKING LTD.

CAPSTONE MANUAL

USER'S MANUAL

Operating instructions for drivers, cashiers, tenant staff, and administrators.

Automated Parking Management System

Group code: CAP-2520-IT

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For day-to-day use of the Parking Ltd. web application and computer vision gate demonstration.

1. Sign in and find your workspace

Open http://127.0.0.1:8080/ while the PHP server is running. Sign in with the account assigned to your role. The app protects pages and API actions by role. Use one browser host consistently: 127.0.0.1 and localhost use different session cookies.

Role	Workspace	Main tasks
Driver	Dashboard, Reserve Slot, Digital Wallet	View availability, register a vehicle, reserve/pay, review citations
Cashier	Cashier POS	Look up a stay, calculate and collect charges, record exit fallback
Tenant staff	Tenant Validator	Validate an eligible Joey's Restaurant dining customer
Administrator	Administration	Manage accounts/VIP, reservations, alerts, fines, slots, audit, sessions

> **Payments in this build**
>
> Reservation checkout is a local demo simulator. It contacts no payment provider, accepts no real payment credential, and moves no money. Cashier cash records are application records for demonstration and audit.

2. Driver: registration and vehicle

1. **Create an account**
   Use Register from the login portal. Provide the required contact, password, and license plate data. Vehicle color is not required at registration; the entrance camera records a color label if the view allows it.
2. **Review availability**
   Open Dashboard or Live Map. Availability is based on the application database and sectional observations. A zone count is not a guarantee that every physical bay was observed; unknown camera bays may remain unverified.
3. **Understand Zone A**
   Zone A is reserved for accounts marked VIP/PWD priority. The system uses a single is_vip priority flag. An administrator can change it in the account directory. If the flag is revoked, non-VIP allocation favors Zone B then the next eligible zone.

3. Driver: reserve a slot

- Open Reserve Slot. Choose the registered vehicle, an available zone or allowed specific bay, and start/end dates. Reservations cover 1 to 3 days in the current system.
- The server calculates the fee. Current fee logic is ₱60 per booked day and a ₱100 surcharge for a reservation spanning at least two days.
- Press Hold Slot & Continue to Demo Checkout. The slot is held for 10 minutes while a pending payment is completed.

- Choose the successful demo outcome, continue to verification, and enter the displayed simulation code. A successful local demo creates a paid reservation and QR pass. Decline, cancellation, and expiry do not create a paid pass.
- Save the reservation QR. The token is linked to the plate and reserved bay; an active paid reservation is used for entry matching. Show it if entrance plate recognition fails.

4. Driver: parking session and QR

For each successful entry, the backend records a PARKED session, assigned slot, entry time, payment state, issued token, and entrance color observation if known. A QR is issued even when the plate is readable. Keep the issued entry QR available for an unreadable exit plate. The QR is an opaque session token; it is not a payment by itself.

5. Cashier: look up and settle

1. **Look up an active vehicle**

   Open Cashier POS. Enter its normalized plate or scan/type its issued entry QR, then choose Query Stay. Review the displayed plate, assigned slot, entrance color, entry time, elapsed duration, charges, and any unpaid fines on earlier sessions for that vehicle.

2. **Take the allowed cashier action**

   Accept the amount shown and record cash received. The tender must cover the total due. All unpaid vehicle citations are itemized and settled with the payment. A lost physical ticket adds a ₱100 exception charge.

3. **Check exit status**

   Use Check by License Plate for a live camera decision or the QR fallback form for an unreadable plate. A valid paid, grace-bypassed, prepaid, or in-window vendor state can clear the digital authorization if identity checks pass.

4. **Record QR exit when needed**

   Query the QR first. Check the displayed session and compare the real vehicle with the entrance color, which is supporting evidence only. Confirm color review when required and submit the issued entry token. The token, not color, authorizes lookup.

Condition	Current charge / action
At or before 15 minutes from entry	₱0 base fee; cashier can record BYPASSED.
After grace, daytime	₱30 flat parking fee.
Stay crosses date or exit is at/after 23:00	₱100 total overnight parking fee.
Tenant validation within 20 minutes	₱30 base fee subsidized; exit allowed in the 20-minute window. After it, ₱30 is due.
Confirmed violation	₱200 per ordinary penalty; settle in Cashier POS before exit.
Lost physical ticket	Additional ₱100. This is distinct from violation penalties.

6. Tenant staff: validate a dining customer

1. **Look up the parked session**

Open Tenant Validator and enter the customer plate or scan the entry token. The session must still be active and parked.

2. **Apply an eligible validation**

   Enter the receipt or invoice reference and submit Apply 20-Min Exemption. The app records the staff identity, receipt reference, vendor timestamp, and ₱30 subsidy in the audit/activity ledger.

3. **Tell the customer the expiry**

   The vendor validation starts a 20-minute departure window. Complete any other due charges or staff review; vendor validation does not waive violation penalties or overnight charges.

7. Administrator: accounts and operations

- Dashboard: review operating counts, revenue summaries, active sessions, and zone occupancy. Check that the data belongs to the current main database.
- User Directory: find a driver and grant/revoke VIP/PWD priority flag. This flag is used for Zone A eligibility; the interface does not maintain separate VIP and PWD categories.
- Active Bookings: review paid reservations and use the available override/cancel controls with an operational reason. Reservation payments are simulated locally in this build.
- Violations Log: inspect sectional alert type, session, details, and evidence image. Confirm only after review to issue a ₱200 ordinary penalty; reject to issue no fine. Lost ticket is the separate ₱100 exception.
- Simulation Sessions: inspect or recover a simulation session that stopped before exit. Use a clear reason. Recovery records an administrative state change; it does not claim that an exit camera observed the car leave.
- Audit Trail and ALPR feed: review timestamped gate events, cashier/vendor actions, reservation and VIP changes, alert decisions, and session closure.
- Slot Overrides: reserved for an administrator. Do not force-free a slot with an open active session; close or recover the session through its session workflow.

8. Computer vision operator controls

Control / mode	Use	Effect
live_vigi.py --detection-only	Check NVR channel connection and recognition	No entry/exit API changes
live_vigi.py	Direct live entrance and exit flow	Channel 1 creates entry; channel 13 requests exit
u key	Unreadable walk-in plate, exactly one vehicle visible	Creates QR-only entry; QR is required at exit
r key	Unreadable plate with an active paid reservation	Prompts for reservation QR token and links entry
full_simulation.py recorded	Recorded gate and sectional workflow	Uses main database and saves evidence/logs
full_simulation.py live-gates	Live VIGI gates plus recorded sectional replay	Uses main database; select an intentionally paired clip

9. Troubleshooting and safe handoff

Message / symptom	Check
No active parked session	Use the current plate or issued QR; check whether it already exited and which database/server is open.
Exit denied: unpaid	Cashier must record parking payment and settle unpaid violations, then replay exit.
Exit denied: review required	Admin must review the pending wrong-slot or Zone A eligibility alert.
Color mismatch	Treat color as a cue for human review. Confirm vehicle identity with entry QR; do not use color alone to authorize exit.
RTSP timeout	Check Mac network route to 192.168.31.2, port 554, channel paths, username, password, and NVR RTSP access.
Plate not confirmed	Inspect focus, lighting, movement, angle, crop, and repeated frames; use QR fallback only for a valid session/reservation case.
Simulation stopped with open session	Resolve payment, alert, tenant, or exit issue. If unrecoverable, Admin > Simulation Sessions records recovery with a reason.

Physical gate responsibility

The current software reports digital exit clearance and records a completed session when a valid camera/QR workflow is accepted. A human attendant must operate the physical barrier and verify passage.